

Kundai Farai Sachikonye
Auf der Lichtung 19, 82178 Puchheim, Germany
github.com/fullscreen-triangle
kundai.sachikonye@bitspark.com
(+49) 1626386344

18 May 2026

Prof. Dr. Astrid Dempfle
Institut für Medizinische Statistik
Universitätsklinikum Schleswig-Holstein (UKSH)
Christian-Albrechts-Universität zu Kiel
Kennziffer 28393

**Re: Wissenschaftlicher Mitarbeiter (m/w/d) — Genetische Epidemiologie /
Statistische Genetik (Kennziffer 28393)**

Dear Prof. Dempfle,

I am applying for the position in genetic epidemiology and statistical genetics at IMS (Kennziffer 28393), in the DFG project on polygenic risk prediction for individuals with elevated familial IBD risk. My background spans genome-wide genomic data analysis validated on large population cohorts, statistical methodology for individual-level disease risk prediction, and multi-omics integration — covering the quantitative and computational requirements the position describes. I note that the position lists fluent German; mine is conversational rather than fluent, and I flag this transparently.

Genomic data analysis at population scale. The Gospel framework processes genome-wide variant data at 10^6 variants per second, integrates RNA-seq quantification alongside genomic data, and applies Bayesian network inference and Gaussian mixture models for multi-omics statistical modelling. Validated against 1000 Genomes Phase 3, gnomAD v3.1.2, and UK Biobank; covers the full pipeline from VCF variant processing through pathway analysis (KEGG, Reactome) to functional annotation. The statistical models — Bayesian networks for genomic variant co-occurrence, variational Bayes for latent component decomposition, GMMs for population structure — are directly applicable to polygenic score construction and familial risk stratification: they provide a probabilistic framework for combining many weak genetic signals into a coherent individual-level risk estimate, which is precisely what the DFG project requires.

Individual-level disease risk prediction and structured risk scores. Beyond standard polygenic score methodology, I have developed a complementary approach to individual disease risk prediction in the Syndrome framework. Disease state is represented as a vector $\mathbf{D} \in [0, 1]^8$ across eight pathophysiological dimensions, each with a per-dimension coherence index $\eta_i \in [0, 1]$ derived from biomarker time-series regularity. The framework locates the patient's current state in a partition geometry on the $[0, 1]^8$ hypercube and computes an $\mathcal{O}(N \log N)$ trajectory to the healthy attractor; therapeutic target selection uses sparse ℓ_1 linear programming to identify the minimum intervention set. For IBD specifically, the eight dimensions map naturally to the distinct pathophysiological components of disease risk — mucosal barrier integrity, immune cell activation, microbial

community structure, stress-response axis, genetic susceptibility load, epigenetic modification state, cytokine balance, and circadian regulatory coherence. A polygenic risk score for IBD captures the genetic susceptibility dimension; the $\mathbf{D} \in [0, 1]^8$ representation integrates it with the remaining seven dimensions into a single structured risk coordinate. This is a methodological contribution to the project's aim of risk prediction in familial settings, where both genetic load and shared environmental factors are elevated: the framework naturally accommodates non-additive interactions between genetic and non-genetic risk components that standard linear PRS cannot represent. A companion paper proves that loop-holonomy self-consistency is the unique viable disease diagnostic criterion for this framework, achieving $\text{AUC} = 1.00$ in 25/25 validation experiments.

Family data and pedigree-based methods. The statistical challenge specific to familial IBD risk prediction is that the units of analysis are correlated — family members share genetic background and often shared environment — which violates the independence assumptions of standard GWAS-derived PRS. My statistical training covers the relevant methodological territory: mixed-effects models for correlated data, Bayesian hierarchical models for shared latent structure, and graph-based methods (Kirchhoff circuit laws on correlation graphs, analogous to kinship matrix eigendecomposition in linear mixed models) for capturing the dependency structure. The Gospel framework's Bayesian network layer handles graphical dependency structures of arbitrary topology, which extends naturally to pedigree graphs.

Programming and statistical software. I work primarily in Python and R. For statistical genetics specifically, I have used the standard genomic data formats (VCF, PLINK bed/bim/fam) and reference databases (gnomAD, 1000 Genomes, UK Biobank) in the Gospel framework. I am familiar with the PRS methodology landscape (clumping and thresholding, Bayesian shrinkage approaches such as LDpred, continuous shrinkage priors) from the literature, though my own implementations use Bayesian network and variational Bayes approaches rather than PLINK-based pipelines. R is my primary language for statistical modelling; Python for data processing and ML pipeline construction.

Background. I hold an MSc in Computational Biology (Universität Tübingen) covering statistical genetics and bioinformatics, an MSc in Pharmaceutical Biotechnology (HAW Hamburg), and a BSc in Biochemistry and Cell Biology (Jacobs University Bremen). I conducted doctoral research in Computational Lipidomics at TUM, covering multi-omics statistical modelling, federated learning for distributed clinical analysis, and large-scale spectral data pipelines. I am legally resident in Germany with full work authorisation. I am available immediately and willing to relocate to Kiel. German is conversational; I am working to improve it actively.

Yours sincerely,

Kundai Farai Sachikonye